



NSA Research Directorate
Deputy Technical Director



NSA @ HackGT

HEARSAY: The Audio Authentication Challenge

VISTA Research, Research Directorate



Text to Speech – Synthetic Speech

Inexpensive and Ubiquitous

- Eleven Labs
 - 10 seconds to 1 minute for Instant Voice Cloning (IVC)
 - Half Hour Plus for Professional Voice Cloning (PVC)
- Amazon Polly
- Grok TTS
- Google Gemini
- Open Weights Models on Hugging Face
 - Qwen-3



Motivation

<https://www.indiatoday.in/world/story/houthis-ai-cloned-yemeni-general-voice-helped-rebels-capture-mocha-middle-east-war-2992703-2026-09-11>

September 11, 2026 | 22:49 IST

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News / World / Houthis in slippers use AI to clone Yemeni Gen's voice, trick troops to withdraw

Houthis in slippers use AI to clone Yemeni Gen's voice, trick troops to withdraw

Preferred source on

The Houthis captured Mocha, a key Yemeni port town on the Red Sea, after deceiving the UAE-backed National Resistance Forces of Yemen to withdraw. For that, the flip-flop-wearing Houthis used an AI-cloned voice message mimicking Yemeni military commander Lt General Tareq Mohammed Abdullah Saleh. This comes after reports of Houthis using Anthropic's Claude to build a ballistic missile.



HEARSAY Challenge

- Given a Set of Audio Files split into training and testing.
 - ~50% real, ~50% synthetic
 - Answers for training are given
 - More than 2 seconds, English only. Some Counter Measures i.e. added noise
 - Added Noise to Real/Synthetic is still Real/Synthetic
 - Hold Out Dataset for Testing
 - No determination of intent. No partially synthetic
- Develop methods to detect synthetic speech
- Definitions
 - Miss – Not identifying Synthetic Speech
 - False Alarm/False Positive – Identifying Real as Synthetic
- False Alarms require analysis. Penalized 4 times greater
- Detection Cost Function (more later)



Submittals

ame	prediction
ame	cm-score
.wav	0.80
.mp3	0.10
.m4a	0.40
.m4a	1.00

- TAB File
 - Two columns, header: File Name, Probability Score (0.0-Real/Bonafide, 1.0-Synthetic/Spoof)

```
TeamName_predictions.tsv
filename      cm-score
file1.wav     0.9
file2.ext     0.1
```
 - No logarithms, log likelihood ratio
 - All Entries
- Submitted to NSA x HexLabs Discord Channel-direct message us your tab-delimited file (.tsv)
- OPTIONAL: One time feedback on .tsv result for testing data.
- FINAL SUBMISSION: Due NLT **8AM 9/27**



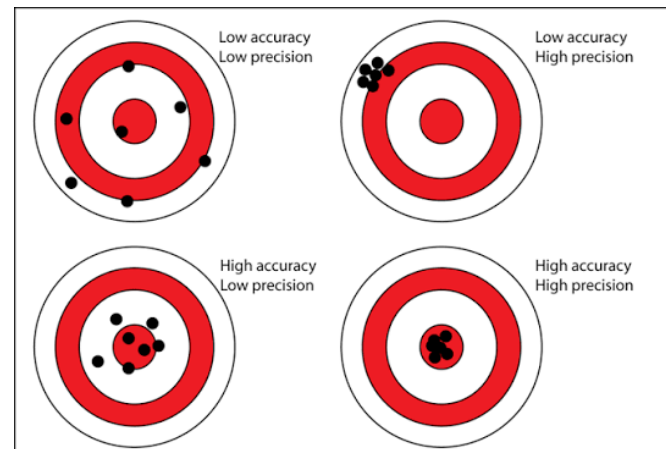
Prizes

- 1st Place – 4 Sonos™ Era 100 Alexa Speaker
- 2nd Place – 4 Audio-Technica™ AT2040 USB Mic
- 3rd Place - 4 JBL™ Flip 5 Bluetooth Speaker
- **Judging Criteria:**
 - 60% Detection Cost Function (MinDCF)
 - 20% Diversity in Techniques – Quality, Innovative, Depth
 - 20% Documentation in GitHub, In your own words



Considerations in Ranking Result

"Far better an approximate answer to the right question, which is often vague, than the exact answer to the wrong question, which can always be made precise." John Tukey (1962), coined term bit, and coinventor of Fast Fourier Transform (Cooley-Tukey). An OG data scientist



https://www.researchgate.net/figure/Precision-versus-accuracy-The-bullseye-represents-the-true-value-eg-the-true_fig6_304674901

False Positives Slow Down Analysis

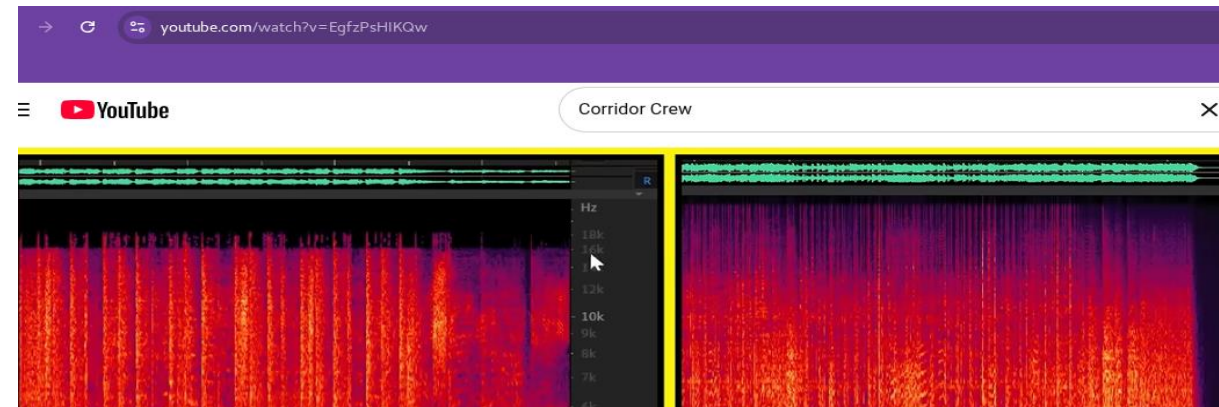
Detection Cost Function

- They may stop work and have to be analyzed. See "I Love Lucy: Chocolate Factory Scene" (YouTube <https://www.youtube.com/watch?v=AnHiAWlrYQc>)
- Cause Distrust - "Boy who cried Wolf"
- Thus we have Detection Cost Function as shown in ASVSpooF V5 (Automatic Speaker Verification)
 - False Positives are penalized by a factor – In our case 4
 - $DCF^* = \text{Cost of Miss} * (1 - P(\text{attack})) * P(\text{Miss}) + \text{Cost_of_False_Alarm} * (P(\text{attack})) * P(\text{FalseAlarm})$ (Minimized)
 - <https://github.com/asvspoof/asvspoof5-baselines> Track 1



What we're looking for

- Unique Perspectives and Approaches
 - Synthetic speech consistency, Breathing, Pauses, Emotions, Prosody
 - Reading Transcript vs Real Time Speech
 - Correlation, Stacking, Signal Processing, Artifacts, Metadata, Statistics
- Test yourself, generate your own synthetic speech
- Tell us what didn't work!
- Tell us what did work and how it can be explained.
 - Minimize Black Box Detectors



YouTube Corridor Crew <https://www.youtube.com/watch?v=EgFzPsHIKQw>



References

- ASV Spoof – <https://www.ASVspoof.org> Automatic Speaker Verification, DCF Definition
- Good illustration of Clipping Artifacts in Debunking AI generated music <https://www.youtube.com/watch?v=EgFzPsHIKQw&t=259s>
- <https://www.youtube.com/watch?v=EgFzPsHIKQw> Corridor Crew debunking AI band
- More to come in Hex Labs Discord Channel!



Tools

- Coding Agents – Claude, ChatGPT, Grok, Qwen
 - VSCode
- FFMPEG – Swiss Army Knife
- ExifTool for Metadata Inspection
- Audacity – Audio Digital Workstation, Exploratory Data Analysis
- Digital Forensics



Audio Sampling Fun Fact

- 44100 Hz, used in CD, DVD, Recording
 - Is multiplied Squares of first 5 primes ($1 \times 1 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7$)
 - Very Factorizable
 - Allowed common factor of NTSC and PAL TV Audio recording.
 - Also more than twice limit of human hearing (20 KHz) due to Shannon-Nyquist Theorem. Required for sampling.
- Beware Aliasing





NSA Presents:

Packet Pursuit: *The Secret File Saga*

*A Packet-Focused Reverse Engineering
Challenge*

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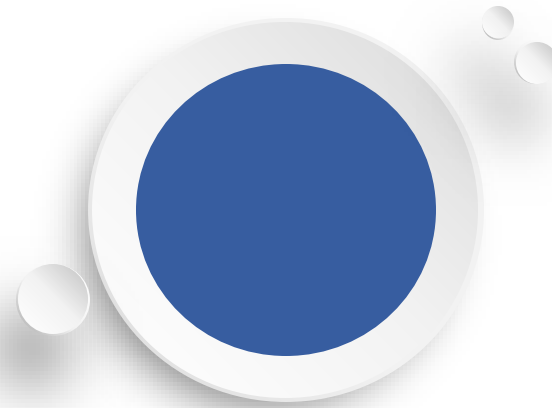
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PART 01

REVERSE ENGINEERING.



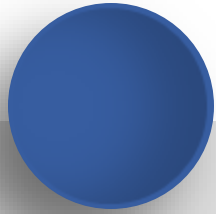
Reverse Engineering

A process or method through which one attempts to understand through deductive reasoning how a previously made device, process, system, or piece of software accomplishes a task with very little (if any) insight into exactly how it does so. It is essentially the process of opening up or dissecting a system to see how it works, in order to duplicate or enhance it. [1][2]

1. [TechTarget.com](https://www.techtarget.com)
2. [Ethics.csc.ncsu.edu](https://ethics.csc.ncsu.edu)

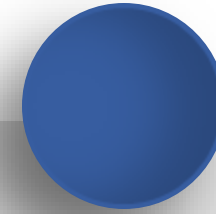


Software Reverse Engineering



Static Analysis

- The analysis of a computer program at rest, meaning not loaded into memory or currently executing.
 - Disassembler: a computer program that translates machine language into assembly language.
 - Examples include Ghidra, IDAPro, WinDbg, Radare2



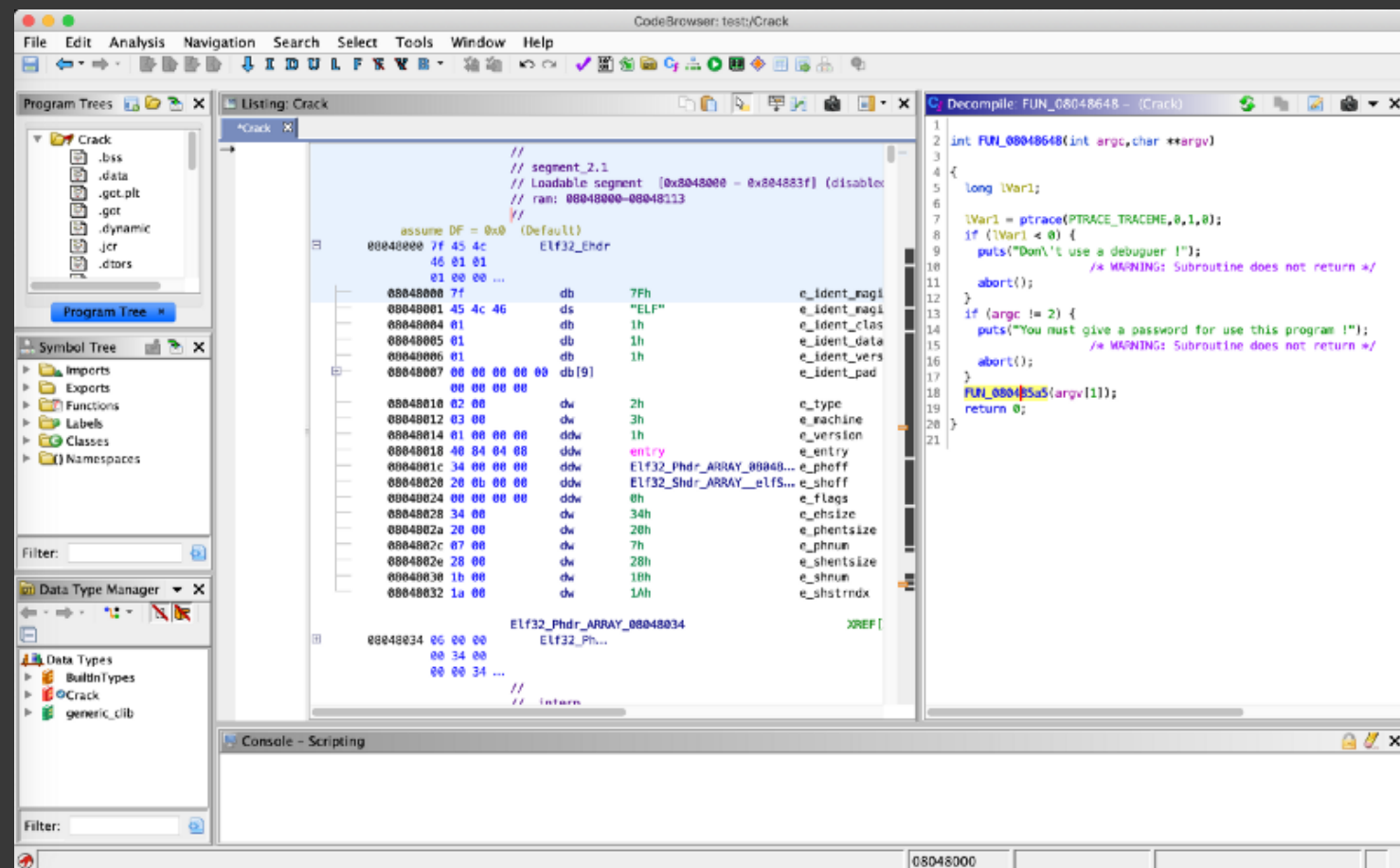
Dynamic Analysis

- The analysis of a computer program that has been loaded into memory and is executing in a controlled environment.
 - Run the program – what happens?
 - Examples include: Radare2, OllyDbg, WinDbg, Valgrind, Java Snoop

GHIDRA

Ghidra is a free, open-source reverse engineering framework developed and maintained by the National Security Agency's (NSA) Research Directorate that allows users to perform static analysis of executables.

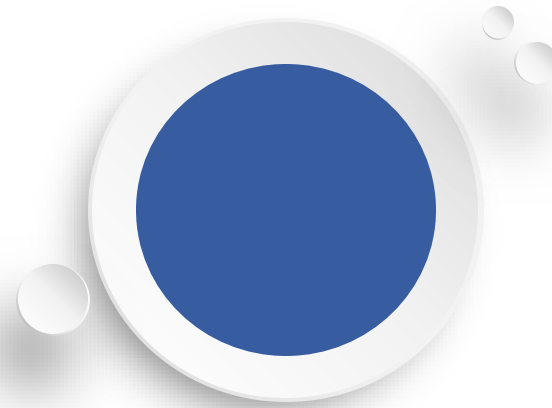
- Disassembles software and handles programs that are compiled for different platforms.
- Decompiles object code back into source code languages like C, C++, Java, or Python.
- Performs Assembly
- Can create function graphs to show how functions interact with each other.
- Allows users to develop their own extension components and scripts using Java or Python.
- Has a built-in graphical debugger that can interact with external debuggers like GDB or WinDbg.





PART 02

NETWORKING



Networking

Computer networking refers to interconnected computing devices that can exchange data and share resources with each other. These networked devices use a system of rules, called communications protocols, to transmit information over physical or wireless technologies.



Network Packets

What?

A network packet is a basic unit of data (bytes) that's grouped together and transferred over a computer network, typically a packet-switched network, such as the internet.

Function

Each packet or chunk of data forms part of a complete message and carries pertinent address information that helps identify the sending computer and intended recipient of the message.

Structure

The size and structure of a network packet are dependent on the underlying network structure or protocol used.

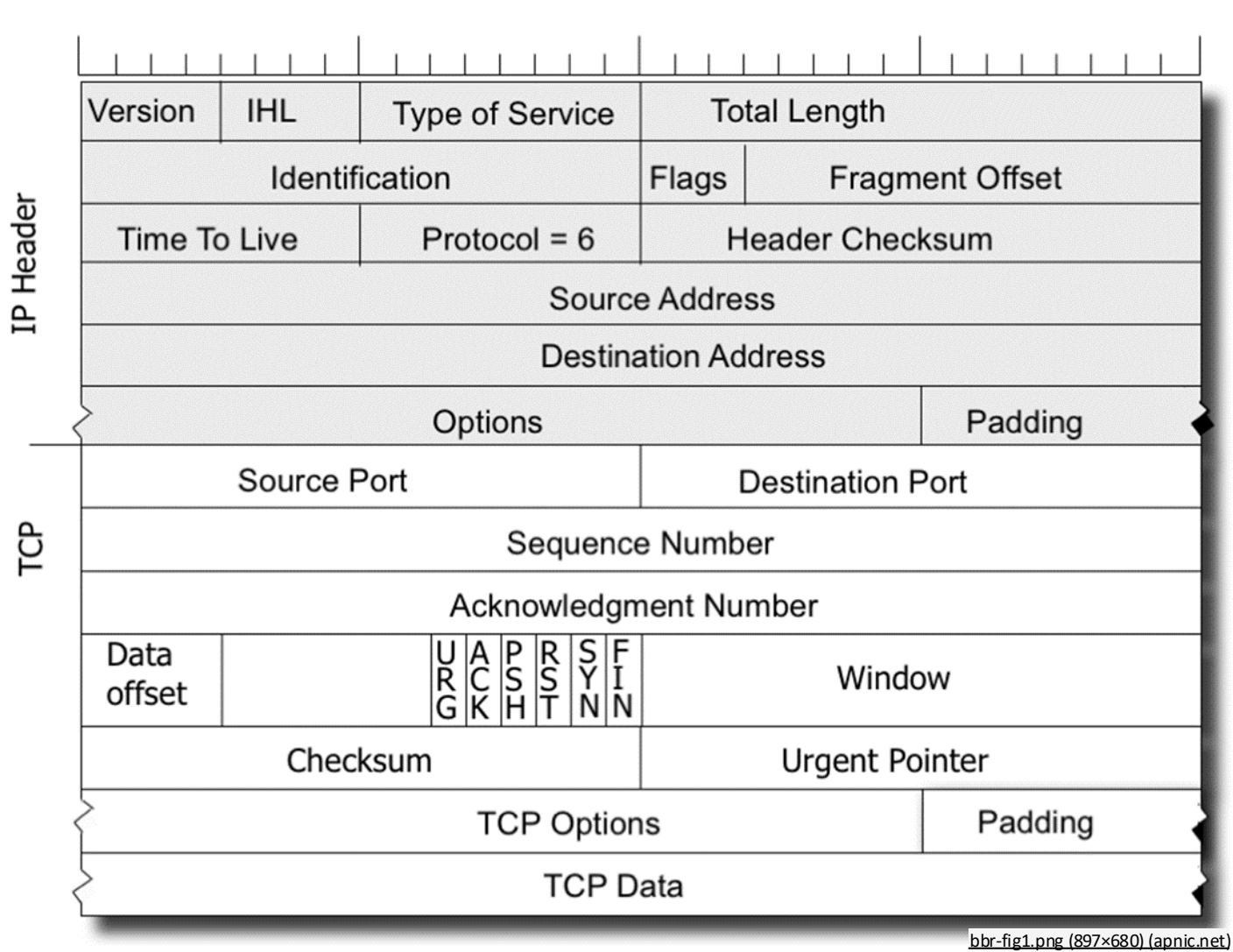
Types

There are many different types of packets: for example: TCP/IP, ICMP, DNS, etc. Each type has its own protocol to be followed.



Example Packet: TCP/IP

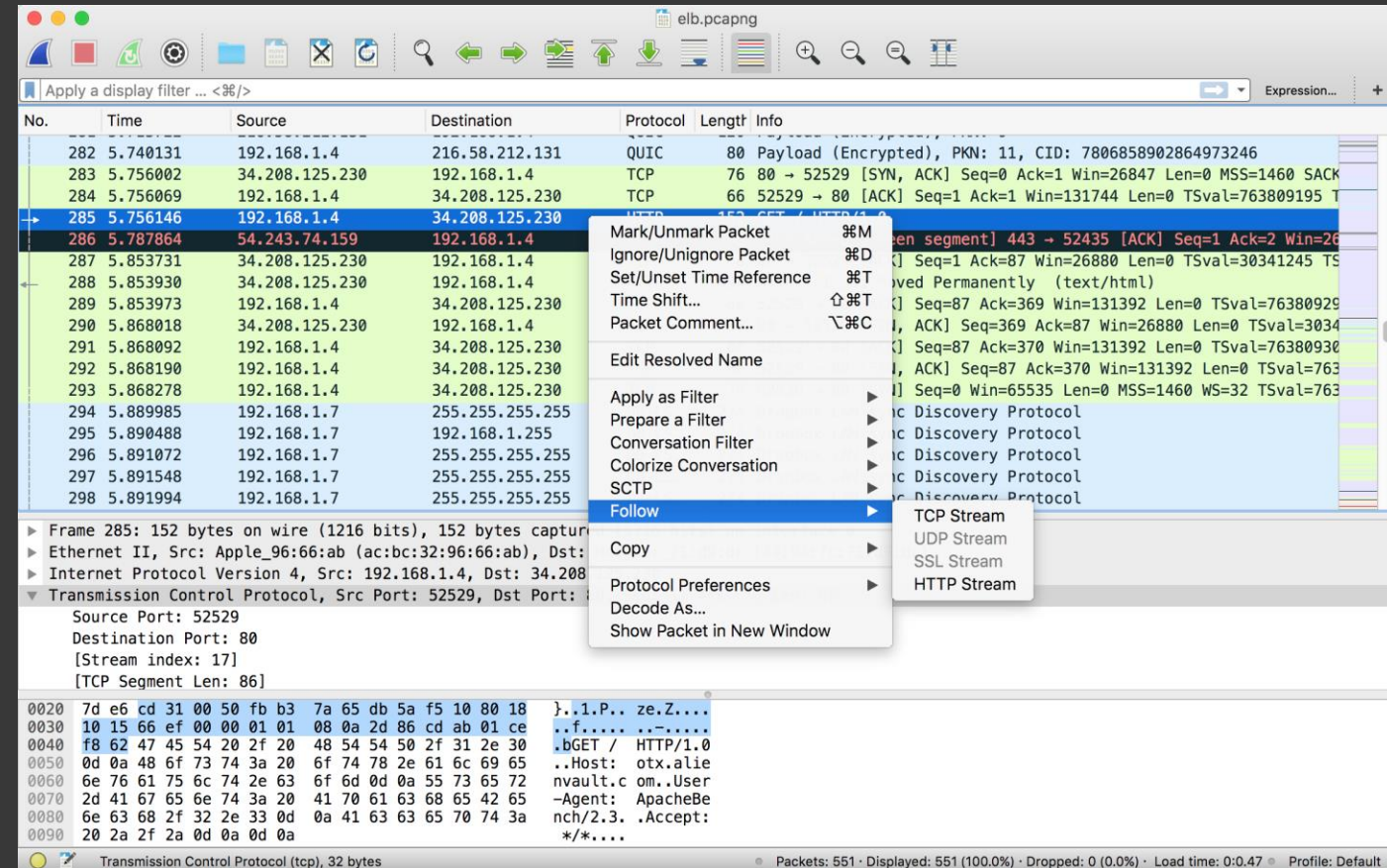
Layers 3 and 4 of the OSI Model



Wireshark

"The world's most popular network protocol analyzer."

- **Packet Capture**
 - Listens to a network connection and capture entire streams of traffic, including tens of thousands of packets at once. It can also read data from a file of already-captured packets.
- **Real-time Analysis**
 - Provides a live view of network traffic.
- **Filtering**
 - Allows users to focus on specific types of network traffic by applying filters to the data.
- **Protocol Understanding**
 - Understands the structure of different networking protocols and parses and display their fields and meanings.
- **Packet Data Display**
 - Displays packets with detailed protocol information and colorizes the display based on filters.
- **Packet Data Manipulation**
 - Allows users to save, export, and search for packet data. It also has tools that can attempt to repair damaged capture files.





PART 03

CHALLENGE.

Scenario

The NSA needs your help! The organization has intercepted network traffic from a group of bad actors who are actively planning an attack. The traffic contains critical information about the attack, and it is up to you to analyze the traffic, discover the hidden information, and reverse engineer the secret file. The security of our nation – and the success of our mission – depends on your expertise and determination.

Packet Pursuit: *The Secret File Saga*

Mission

1. Determine the identity of the malicious group.
2. Locate and reconstruct the hidden file.
3. Decode and RE the file to discover its purpose.
4. Stop the attack.



Scoring Information

Submissions and Scoring via the NSA Dashboard

Part 1

10 Points

PACKET PURSUIT

Part 2

20 Points

PACKET PURSUIT

Part 3

25 Points

PACKET PURSUIT

Part 4

30 Points

PACKET PURSUIT

Part 5

15 Points

PACKET PURSUIT



Prize Information:

The Grand Prize Winner Will Receive:

- An Xbox Series X Gaming Console
- An Xbox Series X Controller
- An Xbox Wireless Headset





PART 04

CONTACT US.

In-Person

Available at the NSA Booth in Klaus Atrium



Contact Us

Discord

Join Our Discord Channel



**Thank
You**

2026 Codebreaker Challenge



Codebreaker Challenge 101

- Annual technical NSA outreach effort.
- Open to U.S. based or U.S. territory schools
- Challenge runs throughout the fall semester (September 2026 – January 2027)



Technical Format

- The participants work through a series of varied difficult tasks designed around a fictional story line. The challenge is looking to enhance critical thinking in real-world scenarios.
- The goal is for students to complete as many tasks as possible, learn about NSA's mission and technical work, and sharpen their cybersecurity skills.



2026 Scenario

- A U.S. Army installation has detected a potential network compromise. The base Security Operations Center (SOC) team has requested NSA's assistance.
- Here is your chance to make an impact by personally defending American interests against a sophisticated and capable adversary.



2026 Tasks

- Welcome to the team, something is not right.
 - To ensure that the challenge is fair to all, we will not be listing the specific tasks for this year's challenge at this moment; however, historically, there have been 6-9 tasks presented.



Skills Learned

- Cryptanalysis
- Forensics
- Network Analysis
- Reverse Engineering
 - Recommended tool is Ghidra
 - Best way to get better is to practice
- Vulnerability Research, Exploitation



Virtualization

- Virtual Machines (VMs) are always used to perform dynamic analysis. Participants should be flexible when it comes to operating systems (Linux or Windows).
- Examples of some virtualization tools include: Vmware, VirtualBox, Hyper-V



Vulnerability Research

- Recommended tool for vulnerability research:
 - Your brain – Learn to think like a hacker, not a developer.
 - Learning the bug classes can assist in your journey.



What's in it for Participants?

- High performers:
 - A letter from NSA
 - NSA fidget spinner
 - Contact from an NSA recruiter regarding potential careers with NSA
- Challenge solvers:
 - A letter from NSA
 - A Codebreaker Medallion (how many is TBD)
 - Contact from an NSA recruiter regarding potential careers with NSA



What's in it for Colleges/Universities?

- Schools will be placed into one of 5 divisions based on the number of registered students. Student scores are added together to determine ranking within each division.
- Division winning schools from divisions 1-3 will earn the coveted Codebreaker trophy.



Getting Started/Reminders

- Get started at <https://nsa-codebreaker.org> and sign up using you school's email address
- See our challenge resource list – <https://nsa-codebreaker.org/resources>
- See <https://nsa-codebreaker.org/thanks>
- Please continue to use the website's "Get help" feature
- Visit <https://www.nsa.gov/careers>
 - *Must be a U.S. citizen
- Learn and have fun!!!!



Thank you!!!

- For general correspondence – codebreaker@uwe.nsa.gov

